FILTER DESIGN TEST CASE

TC-SAW (Temperature Compensated SAW) - Band 30 (2300 MHz) for Millimeter-Wave Beamforming
Design Validation and Performance Characterization

DOCUMENT INFORMATION

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Design Center:	Irvine Design Center
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1. OBJECTIVE

This document describes the design, simulation, and characterization of a TC-SAW (Temperature Compensated SAW) filter targeting Band 30 (2300 MHz) for Millimeter-Wave Beamforming. The filter is designed on Quartz substrate with a target center frequency of 1057.8 MHz and bandwidth of 237.7 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the Millimeter-Wave Beamforming module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: TC-SAW (Temperature Compensated SAW)
- Frequency Band: Band 30 (2300 MHz)
- Application: Millimeter-Wave Beamforming
- Substrate: Quartz
- Package: CSP (Chip Scale Package)
- Design Tool: PathWave EM Design
- Design Center: Irvine Design Center

This specification applies to all variants of the TC-SAW (Temperature Compensated SAW) design targeting Band 30 (2300 MHz). Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency: 1057.8 MHz
Bandwidth (3 dB): 237.7 MHz
Insertion Loss Target: <= 1.3 dB
Return Loss Target: >= 22.0 dB
Out-of-Band Rejection: >= 45.0 dB
Isolation (Duplexer): >= 22.0 dB
Q Factor: 9360
Temperature Coefficient:-17.9 ppm/°C
Die Size: 0.8 x 1.1 mm
Wafer Size: 6-inch
Number of Resonators: 9
Electrode Material: W

Passivation: Si3N4

4. TEST PROCEDURE

1. Open PathWave EM Design and load the TC-SAW (Temperature Compensated SAW) template project.
2. Define the substrate stack: Quartz with specified layer thicknesses.
3. Set design targets: center freq 1057.8 MHz, BW 237.7 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 1.3 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (CSP (Chip Scale Package)).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (6-inch, Quartz).
10. Perform on-wafer probing using Keysight N9041B UXA.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 3173 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 1.3 dB
- Return loss in passband shall be >= 22.0 dB
- Out-of-band rejection at +/- 357 MHz offset >= 45.0 dB
- Group delay variation across passband shall be <= 2 ns
- Temperature drift over -40°C to +85°C shall be <= 2.4 MHz
- Power handling shall be >= +25 dBm at 1 dB compression
- ESD tolerance >= 500 V (HBM)
- Die yield on qualification lot >= 87%

6. TEST RESULTS

Measurement Date: 2024-12-26
Devices Tested: 75
Insertion Loss (meas): 1.01 dB
Return Loss (meas): 24.0 dB
Rejection (meas): 47.6 dB
Isolation (meas): 27.1 dB
Q Factor (meas): 9360
Group Delay Variation: 3.6 ns
Power Handling: +26 dBm
Die Yield: 84.1%

Result Classification: CONDITIONAL PASS

7. OBSERVATIONS AND FINDINGS

a) The TC-SAW (Temperature Compensated SAW) design demonstrated acceptable performance for Band 30 (2300 MHz).
b) Insertion loss met target specification.
c) Out-of-band rejection exceeded the minimum requirement.
d) Slight frequency drift observed at +85°C; TC layer thickness may need adjustment.
e) ESD robustness exceeded specification by 2x margin.

8. CORRECTIVE ACTIONS

1. Review near-band rejection margin for worst-case temperature.
2. Re-run EM simulation with refined mesh for better accuracy.
3. Schedule follow-up measurement after design tweak.

9. SIGN-OFF

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